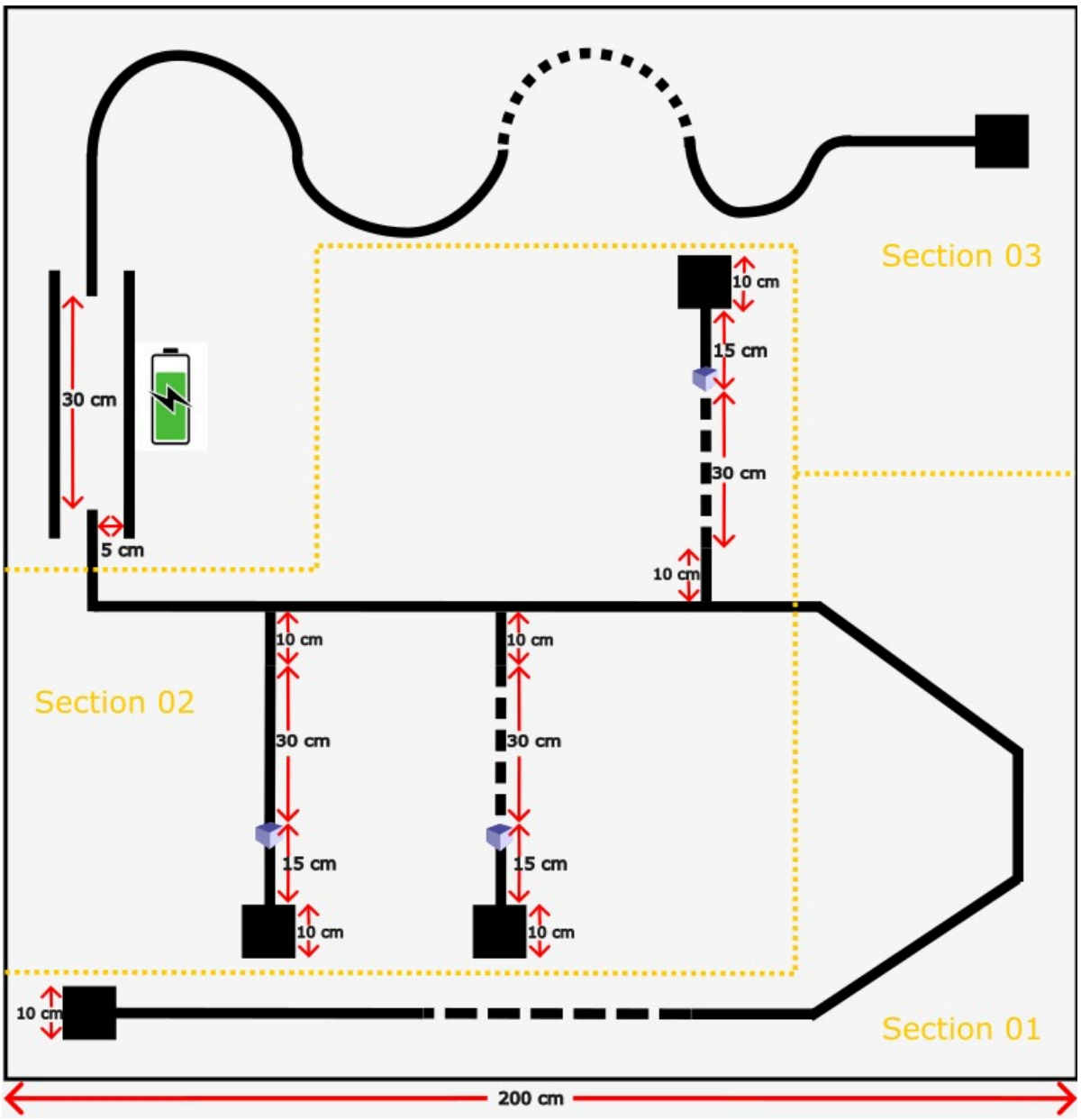
# IESL Robogames - School Category
# Final Task




**Important:** This arena is provided for testing purposes only. The colored markings (red and yellow) shown in the diagram are for illustrative purposes only to indicate different sections and dimensions. These markings will not be present in the actual competition arena.

## 1. Arena Specifications

- Arena size: **2 m × 2 m**
- Background: **White surface**
- Path: **Black line (#000000)**

### Line Properties

- Width: **2 cm**
- Types:
    - **Solid line**
    - **Dashed line**
- Dashed line:
    - Length of a single dash line: **2cm**
    - Gap between dashes(edge to edge): **2 cm**

The path includes:

- Straight segments
- Curves
- T-Junctions

### Black Squares

- Size: **10 cm × 10 cm**
- Used as:
    - Start point
    - End point (GOAL)
    - Cube Placement locations

### Cubes

- Size: **5 cm × 5 cm × 5 cm**
- Material: **White Bristol board**

# 2. Task Description

## 1. Section 1 – Line Following

The robot must start from the start square, with its IR sensor array positioned inside the black square and close to the edge facing the path.

In this section, the robot must follow both **solid and dashed lines**

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# 2. Section 2 – Cube Handling

This section consists of multiple branches connected through T-junctions. Each branch contains a cube that must be handled based on the type of line leading to it.

## 2.1 Path Rule at T - Junctions

At every T-junction, the robot must enter the branch path to handle the cube first. After completing the task, the robot must return to the same junction and continue through the remaining path.

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## 2.2 Branch Structure

There will be exactly **3** branches. Each branch connected to a T-junction is defined as follows:

- A straight line of **10 cm** from the junction
- Followed by a **25 cm segment**, which may be either:
    - Solid line, or
    - Dashed line
- The cube is placed immediately after this segment
- With the cube, there is a **15 cm solid straight line**
- At the end of this segment, a **black square** is placed as the target location

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## 2.3 Cube Handling

When a cube is detected, the robot must act based on the type of line followed immediately before reaching the cube.

**Case 1: Dashed Line (Push Required)**

If the line before the cube is dashed, the robot must:

- Push the cube
- Place it fully inside the black square

After pushing:

- The robot must return to the previous T-junction
- Then continue through the remaining path

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**Case 2: Solid Line (Avoid Required)**

If the line before the cube is solid, the robot must:

- Avoid the cube without any physical contact and turn back

After turning back:

- The robot must return to the previous junction
- Then continue through the remaining path

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# 3. Section 3 – Charging and Final Path

This section includes the charging bay followed by the path leading to the goal.

## 3.1 Charging Area

- Length: 30 cm
- Structure:
    - Two separated black line segments
    - White region in between

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## 3.2 Charging Rules

The robot must:

- Fully enter the white region (entire robot inside)
- Remain inside for a minimum of 5 seconds

The timing is defined as follows:

- The timer starts when the robot fully leaves the previous black line
- The timer ends when the front of the robot reaches the next black line

After charging:

- The robot must reconnect to the black line ahead

Important:

- Side bracket-shaped lines must not be used for line following

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## 4. Goal Condition

After completing all required tasks, the robot must:

- Continue following the dashed and straight lines
- Stop completely inside the final black square (END)

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## 5. Arena Variation (Competition Day)

The arena used in the Final Competition Day will follow the same section structure, but the path layout will be different.

### Section 1 Variations:

- Black line path design may change

### Section 2 Variations:

- Branch position and direction may change
- The type of line (solid or dashed) leading to cubes may change

### Section 3 Variations:

- The path design may include curves and curved dashed lines

### Fixed Constraints:

- Line width, cube size, square size and charging area dimensions will remain unchanged

# 3. Scoring System

# Final Score = Completion Time + Penalties

### Penalties

- Hit cube when on solid line = **+5 seconds**
- Fail to push cube on dashed line = **+15 seconds**
- Incomplete push (cube not fully inside square) = **+5 seconds**
- Skipping required branch = **+20 seconds**
- Charging less than 5 seconds = **+10 seconds**

# 4. Scoring for Incomplete Runs

## Section 1 – Line Following (25 Marks)

- Smooth line following = 15
- Correct handling of dashed lines = 10

## Section 2 – Cube Handling (45 Marks)

Per cube:

- Correct decision (push/avoid) = 10
- Proper execution = 5

Example (3 cubes): 15 × 3 = 45 marks

## Section 3 – Charging + Goal (30 Marks)

- Correct charging behavior = 15
- Proper exit and line reconnect = 5
- Reaching goal = 5
- Proper stop at goal = 5

**Important:** The decision of the judges will be final and binding in all matters related to the competition, including scoring, rule interpretation and penalties and cannot be challenged.

All participants are expected to adhere strictly to the rules and guidelines provided. Any behavior deemed unsafe or unfair may result in disqualification at the discretion of the judges.

****Any updates or modifications to the rules and regulations in this document will be communicated to all participants through official channels. Participants are responsible for staying informed about any such updates.**